

EXPLORATORY DATA ANALYSIS

BASIC ASSIGNMENT

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1)Library Management (List + Loop)

```
books = ["Math", "Science", "English"]
```

```
while True:
```

```
    print("\n1. View Books\n2. Borrow Book\n3. Exit")
```

```
    choice = input("Enter choice: ")
```

```
    if choice == "1":
```

```
        print("Available books:", books)
```

```
    elif choice == "2":
```

```
        book = input("Enter book name: ")
```

```
        if book in books:
```

```
            books.remove(book)
```

```
            print("Book issued")
```

```
        else:
```

```
            print("Not available")
```

```
    else:
```

```
        break
```

```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py

1. View Books
2. Borrow Book
3. Exit
Enter choice: 1
Available books: ['Math', 'Science', 'English']

1. View Books
2. Borrow Book
3. Exit
Enter choice: 2
Enter book name: math
Not available
```

2. Bank System (Functions)

balance = 1000

def deposit(amount):

 global balance

 balance += amount

 print("Deposited:", amount)

def withdraw(amount):

 global balance

 if amount <= balance:

 balance -= amount

 print("Withdrawn:", amount)

 else:

 print("Insufficient balance")

deposit(500)

withdraw(300)

print("Final Balance:", balance)

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_
64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
Deposited: 500
Withdrawn: 300
Final Balance: 1200
PS C:\Users\aliza\Desktop\EDA> █
```

3. Grade Calculator (Loop + Input)

```
marks = []

n = int(input("Enter number of subjects: "))

for i in range(n):

    m = int(input("Enter marks: "))

    marks.append(m)

avg = sum(marks) / n

if avg >= 75:

    grade = "A"

elif avg >= 50:

    grade = "B"

else:

    grade = "C"

print("Average:", avg)

print("Grade:", grade)
```

```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_
64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
Enter number of subjects: 3
Enter marks: 97
Enter marks: 98
Enter marks: 99
Average: 98.0
Grade: A
PS C:\Users\aliza\Desktop\EDA> █
```

4. To-Do List (List + Menu)

```
tasks = []
```

```
while True:
```

```
    print("\n1. Add Task\n2. View Tasks\n3. Remove Task\n4. Exit")
```

```
    ch = input("Enter choice: ")
```

```
    if ch == "1":
```

```
        task = input("Enter task: ")
```

```
        tasks.append(task)
```

```
    elif ch == "2":
```

```
        print("Tasks:", tasks)
```

```
    elif ch == "3":
```

```
        task = input("Enter task to remove: ")
```

```
        if task in tasks:
```

```
            tasks.remove(task)
```

```
    else:
```

```
        break
```

```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
```

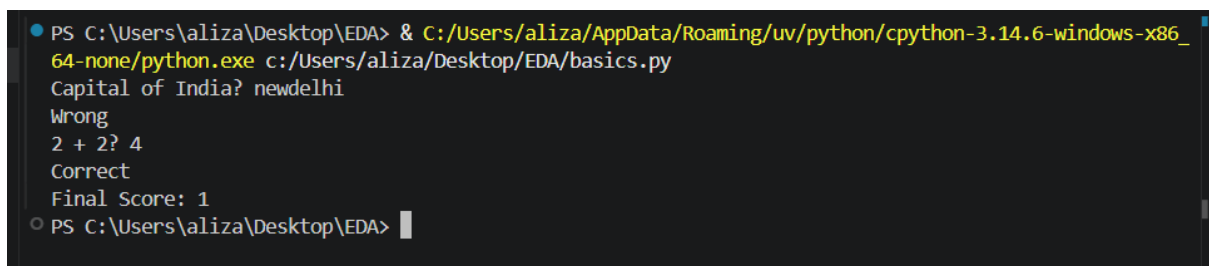
```
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter choice: 1
Enter task: study
```

```
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter choice: 1
Enter task: eat
```

```
1. Add Task
2. View Tasks
3. Remove Task
4. Exit
Enter choice: 2
Tasks: ['study', 'eat']
```

5. Quiz Game (Dictionary + Score)

```
questions = {  
    "Capital of India?": "Delhi",  
    "2 + 2?": "4"  
}  
  
score = 0  
  
for q in questions:  
    ans = input(q + " ")  
    if ans == questions[q]:  
        print("Correct")  
        score += 1  
    else:  
        print("Wrong")  
  
print("Final Score:", score)
```



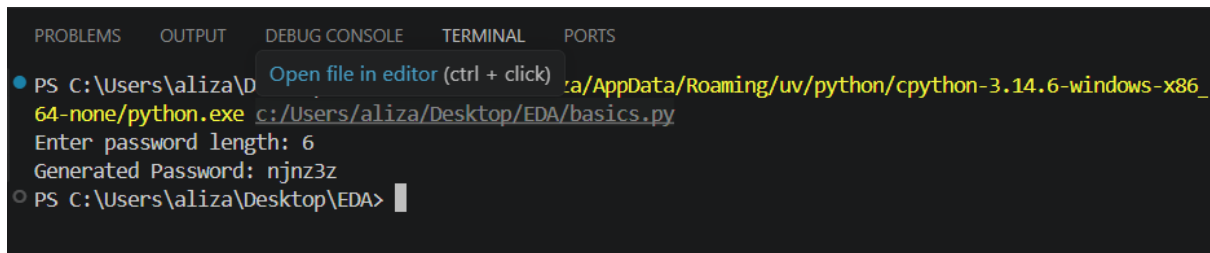
```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py  
Capital of India? newdelhi  
Wrong  
2 + 2? 4  
Correct  
Final Score: 1  
PS C:\Users\aliza\Desktop\EDA>
```

6. Password Generator (Random + User Input)

```
import random  
  
chars = "abcdefghijklmnopqrstuvwxyz123456789"  
  
length = int(input("Enter password length: "))  
  
password = ""  
  
for i in range(length):
```

```
password += random.choice(chars)

print("Generated Password:", password)
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\aliza\D Open file in editor (ctrl + click) :a/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_
64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
Enter password length: 6
Generated Password: njnz3z
PS C:\Users\aliza\Desktop\EDA>
```

7. Guessing Game (While + Hints)

```
num = 7

guess = 0

while guess != num:

    guess = int(input("Guess number (1-10): "))

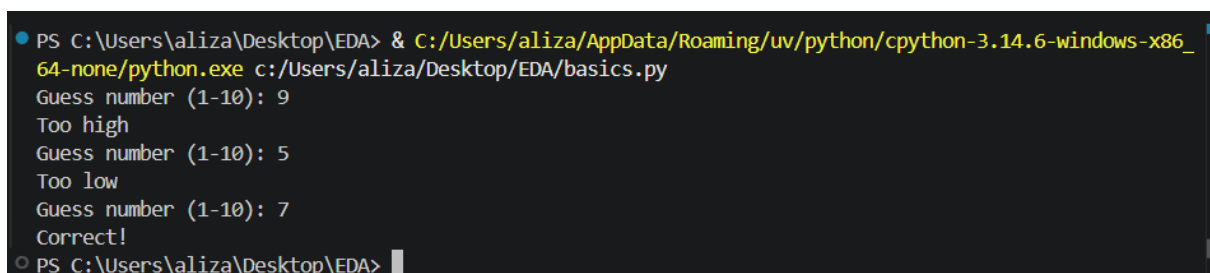
    if guess > num:

        print("Too high")

    elif guess < num:

        print("Too low")

print("Correct!")
```



```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_
64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
Guess number (1-10): 9
Too high
Guess number (1-10): 5
Too low
Guess number (1-10): 7
Correct!
PS C:\Users\aliza\Desktop\EDA>
```

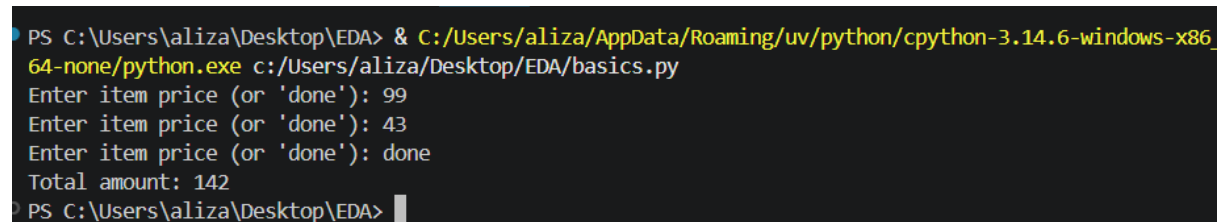
8. Contact Book (Dictionary + Add/Search)

```
contacts = {}  
  
while True:  
    print("\n1. Add Contact\n2. Search Contact\n3. Exit")  
    ch = input("Choice: ")  
    if ch == "1":  
        name = input("Enter name: ")  
        num = input("Enter number: ")  
        contacts[name] = num  
    elif ch == "2":  
        name = input("Enter name: ")  
        print(contacts.get(name, "Not found"))  
    else:  
        break
```

```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py  
  
1. Add Contact  
2. Search Contact  
3. Exit  
Choice: 1  
Enter name: aliza  
Enter number: 46738925537  
  
1. Add Contact  
2. Search Contact  
3. Exit  
Choice: 2  
Enter name: aliza  
46738925537
```

9. Shopping Cart (List + Total)

```
cart = []  
  
while True:  
    item = input("Enter item price (or 'done'): ")  
    if item == "done":  
        break  
    cart.append(int(item))  
  
total = sum(cart)  
print("Total amount:", total)
```



```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py  
Enter item price (or 'done'): 99  
Enter item price (or 'done'): 43  
Enter item price (or 'done'): done  
Total amount: 142  
PS C:\Users\aliza\Desktop\EDA> █
```

10. Attendance Tracker (Set + Input)

```
present = set()  
  
n = int(input("Enter number of students present: "))  
  
for i in range(n):  
    name = input("Enter name: ")  
    present.add(name)  
  
check = input("Enter name to check: ")  
  
if check in present:  
    print("Present")  
  
else:
```



```
print("Absent")
```

```
PS C:\Users\aliza\Desktop\EDA> & C:/Users/aliza/AppData/Roaming/uv/python/cpython-3.14.6-windows-x86_64-none/python.exe c:/Users/aliza/Desktop/EDA/basics.py
Enter number of students present: 4
Enter name: aliza
Enter name: diya
Enter name: tanya
Enter name: jiya
Enter name to check: jiya
Present
PS C:\Users\aliza\Desktop\EDA> █
```